

Understanding and Supporting Autistic Adults in Livestreaming

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ABSTRACT

In my work, I aim to develop opportunities for autistic adults to flourish in online social spaces. I focus on investigating how autistic adults experience social livestreaming platforms such as Twitch.tv and Youtube live so that we can develop tools to support them as streamers. Through interviews with autistic livestreamers, my work provides a first account of the challenges facing autistic livestreamers and also the benefits they experience from streaming, including access to social interactions, gaining control of a public space, and promoting autistic identity. From these findings, I plan to work together with autistic livestreamers using co-design processes to develop tools that improve their experience of livestreaming platforms.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

codesign, autism, livestreaming, livestreamers, Twitch

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1 BACKGROUND AND MOTIVATION

My research is about building opportunities for autistic individuals to flourish—thrive, grow, and achieve—within online platforms built for neurotypical users, but without sacrificing their autistic identity. In my research, I focus on livestreaming sites, including platforms such as twitch.tv and Youtube Live, as a currently understudied social networking service that can support autistic adults' pursuits of representation and social connection. Livestreaming is a form of video-based mixed media where a streamer broadcasts themselves to a live public audience that can interact and react via text chat. While we see examples of autistic users livestreaming today, researchers do not yet clearly understand the benefits and challenges surrounding autistic streaming [17]. Through my work, I will explore the use of livestreaming platforms by autistic adults and develop livestreaming prototypes to support positive social experiences. My research offers a starting point in human-computer

interaction research on autistic adult livestreamers and sheds light on how to support autistic use of online platforms.

Social engagement and support networks are essential to a person's well-being, but these can be difficult to access for those on the autism spectrum. Autistic adults report having a lower quality of life (QoL) that is often linked to social challenges and related co-occurring mental health conditions [15]. Some of the factors that negatively influence the QoL of autistic adults include not having supportive relationships [15], experiencing isolation and social exclusion [18] and masking one's true self during social interactions [6]. This is further exacerbated for autistic adults as they face a "support cliff" as they age out of services and access to social circles they benefited from as children. While autistic children have access to support services and attention from the research community, autism does not stop after childhood.

Past work suggests several ways in which autistic adults can improve their QoL. Mason et al. note that having access to informal, supportive relationships can positively impact QoL for autistics [15]. Cage et al. report that feeling accepted by others as an autistic person may play a part in protecting against depression and that personal acceptance also predicted reduced symptoms of depression [6]. Cooper et al. suggest that having a positive autistic identity could also combat mental health difficulties [8]. However, being on the autism spectrum means significant differences in how someone participates in face-to-face social situations, which can limit in-person access to these benefits. In my research, I investigate how livestreaming platforms as an online social space can fill this gap for autistic adults.

2 RESEARCH OBJECTIVES

The overarching goal of my work is to make online social spaces more inclusive and supportive of autistic adults while allowing those adults to maintain their autistic identity. I move towards this goal by framing my research through three research objectives focused on the context of livestreaming.

- RO1: *Investigate the experience of livestreaming for autistic adults.*
- RO2: *Explore autistic adults' perspectives on the benefits and challenges of livestreaming.*
- RO3: *Develop technology to help autistic adults flourish on livestreaming platforms.*

I formulated these objectives by identifying several under-explored areas of existing autism-related HCI research. One common focus of previous work is on training neurotypical behaviours [13, 19]. This leaves a gap in the current literature on addressing and supporting the strengths, preferences, and goals of autistic individuals. My research investigates how livestreaming can open up opportunities for autistic adults to engage in an activity they enjoy while showcasing their **unique identities and strengths**. Past work in HCI

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has tended to target autistic children [13], and there are many opportunities to study the unique contexts and experiences of autistic adults. For instance, some content creators become self-employed and support themselves monetarily through livestreaming. This has the potential to significantly impact autistic adults, as many are unemployed or underemployed.

Existing research tells us how non-autistic people use livestreaming and the social opportunities provided by livestreaming [9, 11]. Past work has also investigated how different groups use and benefit from building communities through livestreaming, such as those who are blind or have low vision [12], are a part of minority groups [10]. However, we know very little about the autistic experience of livestreaming. My work aims to contribute knowledge about how autistic adults use livestreaming platforms and how we can push toward making livestreaming a **supportive social space** for autistic adult users.

3 RESEARCH APPROACH

My work will take two distinct phases to address the above research objectives.

3.1 Understanding Autistic Adult Livestreamers

My work will, first and foremost, contribute to a holistic understanding of autistic adults' perspectives on livestreaming, including their motivations for streaming and how they manage their streaming experience. This is an essential first step to developing technology for autistic adult livestreamers, as there is no current research on this topic.

I recently completed a qualitative empirical study where I observed and recruited ten adult autistic livestreamers to interview. My preliminary study results have been presented in a poster submission to the ACM SIGACCESS Conference on Computers and accessibility (ASSETS'22) [17]. In my initial findings, I report how the autistic livestreamers I interviewed use livestreaming to practice social interactions, receive social support, build meaningful relationships, and publicly present their autistic identity. After presenting the poster I have been conducting additional reflexive thematic analysis [5] of my interview data. Based on this additional analysis I have a full paper accepted to The International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS'23).

3.2 Co-designing Livestreaming Tools with Autistic Adult Livestreamers

When developing technology for autistic people, it is paramount that we include them in the process whenever possible. Thus, I will run several co-design sessions with autistic adult livestreamers to help design tools and systems to support unique experiences and preferences in livestreaming. Co-design is a development process where the intended audience of a product is included as active participants throughout the development of that product [20].

To help guide the co-design workshops, I will present scenarios and challenges derived from the first phase of my work. Previous co-design studies with autistic co-designers recommend providing concrete narratives and scenarios [4]. I will also offer sketches of potential systems to address some of these concerns. However, my process will focus on getting ideas and feedback from my autistic

co-designers. We will develop a prototype system to build on participants' concerns and ideas from the sessions. I plan to have the co-designers use the prototype we develop in their livestreams so they can provide further evaluation and feedback on it.

Below are two examples of specific experiences drawn from my first phase of work interviewing autistic livestreamers that I will present as ideation starting points for my co-design workshops.

a) Streaming without an audience: Several of the livestreamers I talked to told me that it was difficult to be a new livestreamer as they had to stream without an audience. Streamers are expected to be entertaining and talkative even when very few people or nobody is watching. To address this, I will engage with co-autistic adult livestreamers as co-designers to help determine what tools would have benefited them when they first began livestreaming.

I suggest a potential prototype here to probe what might be possible. I will use examples like this to structure my co-design sessions with autistic streamers by presenting them as scenarios, mockups, and sketches. However, these will serve only as a starting point, and the systems we develop will be based on my co-designers' thoughts and ideas.

Example Probe: In prior work, I addressed the scenario of streaming without an audience—where streamers need to maintain their performance even while nobody is watching. I provided novice streamers with a simulated audience to encourage them to interact despite having no viewers [16]. I developed that prototype before my focus on autistic adults. However, to ensure such a system would be helpful to autistic adults, we must consider adapting it to their communication preferences. For instance, many autistic adults face difficulty engaging in “small talk” [18] and may prefer a simulated audience to avoid this type of conversation. Beyond this specific example, our co-design session could more broadly address how to develop onboarding tools to assist autistic adults in livestreaming as newcomers.

b) Supporting Autistic Livestreamer Goals: We will focus on designing tools to promote the aspects of streaming that autistic adults find beneficial. For example, for some streamers I interviewed, livestreaming positively impacted their in-person social interactions because they could leverage their stream to practice social skills. Needing to engage with an audience meant that streamers were required to interact with a variety of people about a range of topics.

Example Probe: We could create a livestreaming tool to make this ad-hoc skill acquisition more systematic. The system could encourage a streamer and their audience to discuss topics outside the streamer's special interests. Our system would provide an interface integrated into the livestream where the streamer or audience members could add topics to discuss. This would be beneficial because it separates the tracking of discussion topics from the regular scrolling chat interaction. In the typical chat, old messages are pushed out of view as new messages appear. Our work in developing this type of technology promises to make livestreaming a more effective platform to provide benefits that autistic adults gain through streaming.

4 RESULTS AND CONTRIBUTIONS TO DATE

Background work on livestreaming: Before focusing on autistic adults, I participated in researching other areas related to livestreaming, such as mukbang (eating performance) [3] and online teaching

of hands-on activities [14]. I also developed a simulated audience chatbot to assist new livestreamers with performing [16]. Though these do not address autistic adults, they have contributed to my understanding of how people use livestreaming for different purposes and helped to inform my proposed research.

First-hand accounts of autistic adult livestreamers: For my first phase of work, I have already conducted an interview study to gather first-hand accounts of autistic adult streamers' experiences. My preliminary study results have been accepted in a poster submission to the ASSETS conference [17]. Initial findings from my interviews show that the unique mix of media-rich and text-based interactions in a live format offers autistic streamers several benefits. For some, streaming fulfills a desire to build social connections unavailable to them through typical in-person means. Others streamed to practice social interactions where they could be in control of the space—streamers can revoke audience members' ability to chat. And some enjoyed the mixed-media format that allowed them to share their true selves and be publicly visible autistic adult role models.

5 CONCLUSION

My work aims to significantly impact autistic adults' access to peers and community through livestreaming. Livestreams have been described as a digital "third place"—a meeting space outside the home ("first place") and work ("second place"), where people can socialize amongst their peers as one would in a café or at church [11]. An estimated 64% of autistic adults in Canada are unemployed [7]—they have no "second place". Furthermore, while participating at an in-person "third place" such as a café might seem straightforward to someone without autism, those interactions may be difficult or impossible for many autistic adults. Adapting livestreaming for autistic adults can give socially disenfranchised people access to an otherwise inaccessible and valuable "third place". While past research has investigated the benefits and challenges for other minority and marginalized groups on livestreaming platforms, there has yet to be the same consideration for autistic livestreamers.

Lastly, another goal of my work is to advocate for the visibility and representation of autistic adults in broader society through livestreaming. There has been a growing positive portrayal of autistic adults in popular media such as in *Extraordinary Attorney Woo* [2]. However, shows like this and others tend to be built around autistic savants—the protagonist has a unique genius in a field such as medicine or law. This leaves an underrepresented majority of non-savants that society does not see. Furthermore, though the actors in these series have been praised for their portrayal of autistic characters, the actors are not autistic. Fortunately, this is changing. *As We See It* [1] depicts the everyday lives of three diverse autistic young adults played by autistic actors. This series helps the world see the day-to-day challenges and unique strengths of autistic adults. In my research I hope to build on this by enabling more autistic people to showcase themselves to society through livestreaming.

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